

Problem 7.16

If we are promised \$1,500 5 years from now, what is its present value today at:

- (a) 7% simple interest?
- (b) 7% compound interest?

Solution

(a) Simple interest

Under simple interest,

$$1500 = X(1 + it),$$

with $i = 0.07$ and $t = 5$. Hence,

$$1500 = X(1 + 0.07 \cdot 5) = 1.35X.$$

So,

$$X = \frac{1500}{1.35} \approx 1111.1111.$$

Therefore, the present value is

$$\boxed{\$1111.11}.$$

(b) Compound interest

Under compound interest,

$$X(1.07)^5 = 1500.$$

Thus,

$$X = \frac{1500}{(1.07)^5} \approx 1069.7922.$$

Therefore, the present value is

$$\boxed{\$1069.79}.$$